

Number Systems

沈威宇

June 9, 2026

Contents

1 Number Systems

I Numeral system

A numeral system is a set of symbols, called numerals, and the rules for using them to represent numbers. Numerals can be used for counting (as with cardinal numbers), for ordering (as with ordinal numbers), labels (as with telephone numbers), and for codes (as with ISBNs). In common usage, a numeral is not clearly distinguished from the number that it represents.

II Number system

A number system is a theory, proof system, or interpretation structure whose variables are called numbers.

III Number set

A number set is a set of number.

IV Natural numbers (自然數), counting numbers, or whole numbers

The positive integer system (aka the positive integers) $\mathbb{N}^*$, $\mathbb{N}_1$, $\mathbb{N}^+$, or $\mathbb{Z}^+$ is, 1, 2, 3, and so on

The non-negative integer system (aka the non-negative integers) $\mathbb{N}^0$ or $\mathbb{N}_0$ is, 0, 1, 2, 3, and so on.

The natural/counting/whole number system (aka the naturals) $\mathbb{N}$ is defined as either the positive integer system or the set of non-negative integer system; if not specified, $\mathbb{N} = \mathbb{N}_1$ is used.

i Second-order Peano Axioms (皮亞諾公理) (PA_2), Dedekind–Peano axioms, or Peano postulates (皮亞諾公設)

Second-order Peano Axioms (aka Dedekind–Peano axioms or Peano postulates) (PA_2) is a formal construction of the natural numbers. Below, we define $\mathbb{N}_0$; the definition of $\mathbb{N}_1$ can be constructed by simply replace 0 with 1 in the following axioms.

- 0 is a natural number.
- Every natural number n has a successor $S(n)$ which is also a natural number, that is, the natural numbers are closed under S .
- 0 is not a successor of any natural numbers, that is,

$$\forall n \in \mathbb{N}_0 : S(n) \neq 0.$$

- S is an injection, that is,

$$\forall m, n \in \mathbb{N}_0 : S(m) = S(n) \implies m = n$$

- The (second-order) induction axiom: If K is a subset of $\mathbb{N}_0$ such that, 0 is in K , and, for every $n \in \mathbb{N}_0$, $n \in K \implies S(n) \in K$, then $\mathbb{N}_0 = K$.

ii Peano Arithmetic (皮亞諾算術)

The axiomatization of arithmetic provided by Peano axioms.

- **Addition:** Addition is a function $\mathbb{N}_0^2 \rightarrow \mathbb{N}_0$ defined recursively as

$$a + 0 = a,$$

$$a + S(b) = S(a + b).$$

- **Multiplication:** Multiplication is a function $\mathbb{N}_0^2 \rightarrow \mathbb{N}_0$ defined given addition recursively as

$$a \cdot 0 = 0,$$

$$a \cdot S(b) = a + (a \cdot b).$$

- **Preservation of order under addition and multiplication:** $\mathbb{N}$ is a totally ordered set, that is, it is equipped with a transitive relation $\leq$ that is strongly connected and antisymmetric.

- Preservation of order under addition:

$$\forall x, y, z \in \mathbb{N}_0 : x \leq y \implies x + z \leq y + z.$$

- Preservation of order under multiplication:

$$\forall x, y \in \mathbb{N}_0 : 0 \leq x \wedge 0 \leq y \implies 0 \leq xy.$$

- **Strong order:** The strong order $<$ over $\mathbb{N}_0$ is defined as:

$$\forall x, y \in \mathbb{N}_0 : x < y \iff (x \leq y \wedge x \neq y).$$

iii First-order Peano Axioms (PA)

A weaker first-order formal construction of $\mathbb{N}_0$ can be obtained by explicitly adding the addition and multiplication operation and replacing the second-order induction axiom with a first-order axiom schema, called first-order Peano Axioms (PA).

- 0 is a natural number.
- Every natural number n has a successor $S(n)$ which is also a natural number, that is, the natural numbers are closed under S .
- 0 is not a successor of any natural numbers, that is,

$$\forall n \in \mathbb{N}_0 : S(n) \neq 0.$$

- S is an injection, that is,

$$\forall m, n \in \mathbb{N}_0 : S(m) = S(n) \implies m = n$$

- **Addition:** Addition is a function $\mathbb{N}_0^2 \rightarrow \mathbb{N}_0$ defined recursively as

$$a + 0 = a,$$

$$a + S(b) = S(a + b).$$

- **Multiplication:** Multiplication is a function $\mathbb{N}_0^2 \rightarrow \mathbb{N}_0$ defined given addition recursively as

$$a \cdot 0 = 0,$$

$$a \cdot S(b) = a + (a \cdot b).$$

- **Preservation of order under addition and multiplication:** $\mathbb{N}$ is a totally ordered set, that is, it is equipped with a transitive relation $\leq$ that is strongly connected and antisymmetric.

- Preservation of order under addition:

$$\forall x, y, z \in \mathbb{N}_0 : x \leq y \implies x + z \leq y + z.$$

- Preservation of order under multiplication:

$$\forall x, y \in \mathbb{N}_0 : 0 \leq x \wedge 0 \leq y \implies 0 \leq xy.$$

- **The first-order induction axiom (schema):** For any formula $\varphi(x, \bar{y})$ where $\bar{y}$ stands for $y_1, y_2, \dots, y_k$ in the language of Peano arithmetic (i.e. addition and multiplication),

$$\forall \bar{y} : (\varphi(0, \bar{y}) \wedge \forall x : \varphi(x, \bar{y}) \implies \varphi(S(x), \bar{y})) \implies \forall x : \varphi(x, \bar{y}).$$

V Theories Strictly Weaker than PA

i PA^-

PA^- is PA minus the induction axiom.

ii Robinson Arithmetic (Q)

Robinson Arithmetic (Q) is a first-order theory that is strictly weaker than PA^- in the language $L = \{0, S, +, \cdot\}$, where 0 is a constant, $S(x)$ denotes successor of x , $x + y$ denotes addition, and $x \cdot y$ denotes multiplication. Its axioms are:

$$S(x) \neq 0.$$

$$S(x) = S(y) \implies x = y.$$

$$y = 0 \vee \exists x : S(x) = y.$$

$$x + 0 = x.$$

$$x + S(y) = S(x + y).$$

$$x \cdot 0 = 0.$$

$$x \cdot S(y) = x \cdot y + x.$$